

Drug Target Interaction

DRUGS
Chemicals of low molecular masses (~100-500u) and produce a biological response

RECEPTORS
Proteins for communication system in the body

ENZYME INHIBITORS
Drugs that block the binding site of the enzyme and prevent the binding of substrate.

COMPETITIVE INHIBITORS
Drugs compete with natural substrate for their attachment on the active sites of enzymes.

ALLOSTERIC SITE:
Some drugs do not bind to the enzyme's active site. These bind to a different site of enzyme.

CHEMICAL MESSENGERS
In human body, message between two neurons and the neurons to muscles is communicated through certain chemicals. These are called messengers.

ANTAGONISTS
Drugs that bind to the receptor site and inhibit its natural function

AGONISTS
Drugs that mimic the natural messenger by space on the called agonists.

Therapeutic Actions of Drugs

Antacid

Neutralise stomach acids
 • Sodium hydrogencarbonate
 • Aluminium hydroxide
 • Magnesium hydroxide
 • Magnesium carbonate
 • Cimetidine } Reduce secretion of pepsin & HCl in stomach due to histamine
 • Ranitidine

N.A.D

(Neurologically active drugs)
 Tranquilizers and analgesics are neurologically active drugs. These affect the message transfer mechanism from nerve to receptor.

Antihistamines

Anti allergic & antacid

• Diphenylhydramine
 • Chlorpheniramine
 • Brompheniramine
 • Promethazine
 • Terfenadine

Tranquilizers

Treatment for stress & mental diseases

• Nembutal
 • Chlordiazepoxide
 • Meprobamate
 • Equanil
 • Veronal
 • Amytal
 • Luminal
 • Seconal
 • Valium
 • Serotonin

Antifertility Drugs

(Synthetic progesterone derivatives)
 progesterone suppresses ovulation

• Norethindrone
 • Novestrol
 • Mestranol
 • Mifepristone

Analgesics

Pain killers
 • Aspirin
 Relief skeletal pain due to arthritis
 Antipyretic
 Antiblood clotting agent (prevention of heart-attack)

• Paracetamol
 • Novalgin
 • Phenacetin
 • Brufane
 • Morphine
 • Heroin
 • Codeine

Non narcotic

Non addictive
 • Aspirin
 • Paracetamol

Narcotic

Addictive
 • Morphine

Antiseptics

on living tissues
 • Furacine
 • Soframicine
 • Chloroxylonol } Dettol
 • Terpineol
 • Tincture of iodine (2-3% solution in alcohol-water mixture)
 • Iodoform
 • Bithional (In soaps)
 • 0.2% Phenol
 • Iodine (Powerful)
 • Boric acid (Weak antiseptic for eyes)

Antimicrobials

• An antimicrobial destroy prevent development or inhibit the pathogenic action of microbes such as bacteria, fungi, virus, parasites selectively
 • Antibiotics antiseptics and disinfectants are antimicrobial drugs

Disinfectants

Either kill or prevent the growth of microorganisms
 • 1% Phenol
 • DDT
 • 0.2-0.4 ppm solution of chlorine
 • SO₂ in very low concentration

Antibiotics

On basis of mode of control of microbial diseases

Bactericidal

Penicillin
 Aminoglycosides
 Ofloxacin

Bacteriostatic

Erythromycin
 Tetracycline
 Chloramphenicol

Salvarsan (arsenic based) used for treatment of syphilis

On basis of mode of its spectrum of action

Broad spectrum

Ampicillin
 Amoxycillin } Synthetic modifications of penicillins
 Chloramphenicol
 Chloramycetin
 Tetracycline
 Vancomycin
 Ofloxacin

Narrow spectrum

Penicillin G
 Clindamycin
 Metronidazole

Limited spectrum

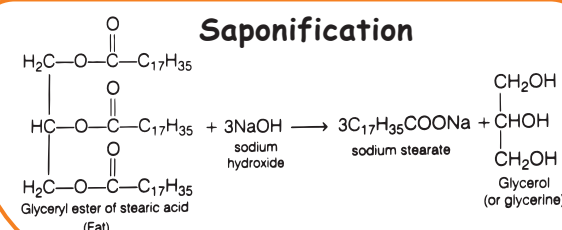
Dysidazine

Soaps & Detergents

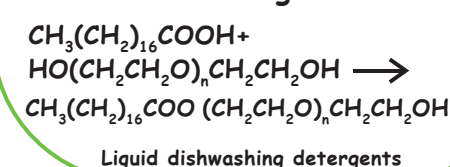
Soaps

Sodium and Potassium salts of higher fatty acids (carbon atoms 12 or higher)

Saponification



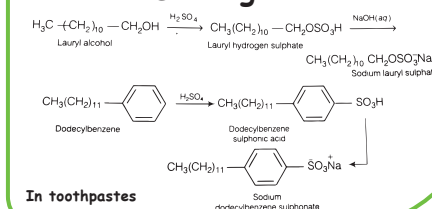
Non-ionic Detergents:



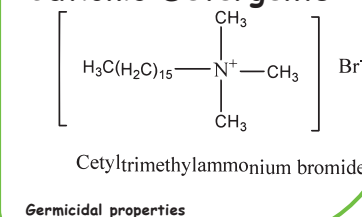
Detergents

Are sodium and potassium salts of long chain sulphonic acids

Anionic Detergents:



Cationic Detergents:



Chemicals in foods

(i) Artificial sweetening agents:

Artificial Sweetener	Sweetness value in comparison to cane sugar
Aspartame (Widely used, unstable at cooking T)	100
Saccharin	550
Sucralose (stable at cooking T, Does not provide calories)	600
Alitame (Stable than Aspartame, High potency sweetener)	2000

(ii) Food preservatives:

Prevent spoilage of food due to microbial growth. Eg: Table salt, sugar, vegetable oils, C₆H₅COONa, Salts of sorbic acid & propanoic acid

(iii) Antioxidants in Food

Butylated hydroxytoluene (BHT), butylated hydroxy anisole (BHA)-Increases shelf life of butter from months to years, SO₂ & sulphite-In beer & sugar syrups